Gradient Descenders at DialectSentEval 2026: A Comparative Study of Pretrained Arabic Transformers for Multi-Dialect Sentiment Analysis

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Abstract

This paper presents the Gradient Descenders system for DialectSentEval Subtask 1 on multi-dialect Arabic sentiment analysis. Because general-purpose models often miss dialectal nuances, we compare three pretrained encoders with distinct linguistic orientations: MARBERTv2, DarijaBERT, and AraBERT. We formulate the task as three-class classification, employing a parameter-efficient head-only adaptation strategy that freezes encoder parameters. Our results show MARBERTv2 outperforms the others, achieving a Macro F1 of 80.11% and Accuracy of 81.90%, securing fifth place overall. These findings emphasize that backbone selection is critical; specifically, broad dialect-oriented pretraining yields the most transferable representations for multi-dialect generalization.

1 Introduction

Arabic sentiment analysis is significantly complicated by the linguistic diversity of regional dialects. Unlike Modern Standard Arabic (MSA), dialectal Arabic exhibits considerable variations in vocabulary, morphology, syntax, and informal expressions, often alongside weakly standardized written forms. These differences hinder sentiment classification systems from generalizing consistently, as sentiment-bearing expressions and their surface realizations vary widely across regions. Consequently, recent work increasingly leverages dialect-aware pretrained language models and multi-dialect evaluation settings. This motivates the development of robust systems capable of handling heterogeneous Arabic varieties rather than optimizing for a single dialect.

DialectSentEval 2026 addresses these challenges through a shared task focusing on Arabic sentiment analysis and swapping (Ezzini et al., 2026a). We participate in Subtask 1: Arabic Dialect Sentiment Analysis, formulated as a three-class classification problem where inputs are labeled positive, negative, or neutral. This task utilizes the Multi-Dialect-Sent (MDS-3) dataset (Ezzini et al., 2026b), which comprises 3,000 sentences spanning five Arabic dialects: Darija, Egyptian, Lebanese, Jordanian, and Saudi. The official evaluation employs the Macro F1-Score as the primary metric, thereby encouraging systems to maintain balanced predictive performance across sentiment classes instead of favoring frequent labels.

In this paper, we investigate the effectiveness of fine-tuning Arabic pretrained encoders for multi-dialect sentiment classification. We specifically experiment with three Arabic pretrained encoders (Abdul-Mageed et al., 2021; Gaanoun et al., 2023; Antoun et al., 2020) that provide varying degrees of alignment with dialectal and general Arabic text (Inoue et al., 2021). Our study offers two primary contributions: (i) we evaluate multiple Arabic pretrained encoder models under a uniform fine-tuning setting on the MDS-3 dataset; and (ii) we comprehensively compare their performance using the official Macro F1-Score alongside Accuracy to analyze differences among the resulting systems.

2 Background

2.1 Benchmark Construction

Arabic sentiment analysis relies on annotated benchmarks that initially focused on specific varieties or domains. Early resources like ASTD and ArSenTD-Lev established foundational evaluation settings (Nabil et al., 2015). Recent shared tasks, such as NADI (Abdul-Mageed et al., 2024) and AHaSIS (Alharbi et al., 2025b,c), have shifted toward standardized multi-dialect evaluations. AraSentEval 2026 extends this via the unified MDS-3 benchmark (Ezzini et al., 2026b). Historically, inconsistencies in dialect coverage, do-

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mains, and label spaces have complicated cross-study comparisons, underscoring the critical need for unified multi-dialect evaluations.

2.2 Lexicon and Feature-based Approach

Early Arabic sentiment systems heavily utilized sentiment lexicons (e.g., SLSA, NileULex) to capture polarity-bearing colloquialisms (Mohammad et al., 2016; El-Beltagy, 2016). These were commonly combined with handcrafted features (such as n-grams, TF-IDF, or embeddings) and traditional classifiers like SVMs or Naive Bayes. While these approaches demonstrated the utility of Arabic-specific lexical data, their effectiveness relies strictly on lexicon coverage and explicit feature design. This dependence restricts their scalability in multi-dialect settings, where extensive lexical and orthographic variations hinder cross-dialect transferability.

2.3 Language Model Approach

Arabic pretrained Transformers shifted sentiment classification toward contextual learning (Alharbi et al., 2025a). AraBERT established general Arabic pretraining for downstream tasks (Antoun et al., 2020). Later models incorporated broader linguistic variations: MARBERT and MARBERTv2 emphasized dialectal social-media data, while CAMeLBERT encompassed Modern Standard, Classical, and dialectal Arabic (Abdul-Mageed et al., 2021; Inoue et al., 2021). Concurrently, dialect-specialized encoders emerged, such as DarijaBERT for Moroccan Arabic (Gaanoun et al., 2023). These advances indicate that pretraining scope significantly impacts multi-dialect performance. However, varying datasets and protocols across existing studies prevent direct comparisons. Consequently, our work evaluates general, broadly dialect-oriented, and dialect-specialized encoders uniformly under the MDS-3 fine-tuning setting.

3 Dataset Analysis

Figure 1 shows a largely balanced distribution across both dialects and sentiment classes, with an important difference between the training and test sets. The training data contain comparable numbers of positive, negative, and neutral instances for Darija, Egyptian, Jordanian, and Saudi, while Lebanese is absent from the training split. In contrast, the test set is evenly distributed across all five dialects, including Lebanese. The main takeaway is therefore that, although class and dialect imbalance is limited for the observed training varieties, the evaluation additionally requires the models to generalize to Lebanese as an unseen dialect, making cross-dialect transfer an important aspect of system performance.

Table 1: Overview of the pretrained encoder architectures. Tok: Tokenization method; SP: SentencePiece; WP: WordPiece. Param: Number of parameters.

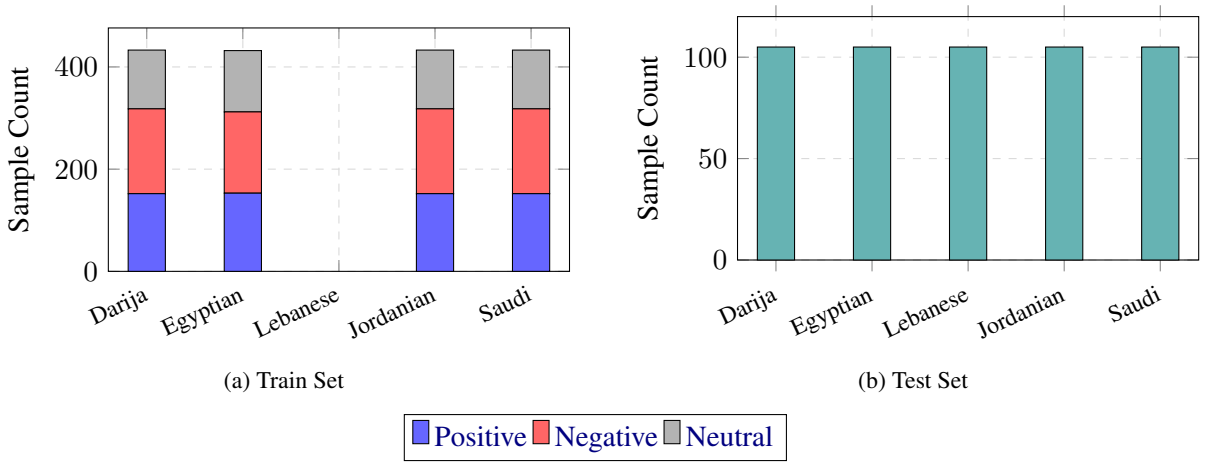
Model	Vocabulary		Param
	Tok	Size	
MARBERTv2	WP	100K	163M
DarijaBERT	WP	80K	100M
AraBERT	SP	64K	136M

4 System Overview

Our system formulates Arabic dialect sentiment analysis as a three-class sequence classification problem. Given an input sentence, the system predicts one of three sentiment labels: positive, negative, or neutral. We investigate three pretrained Arabic encoder models—MARBERTv2 (Abdul-Mageed et al., 2021), DarijaBERT (Gaanoun et al., 2023), and AraBERT (Antoun et al., 2020)—which differ in the linguistic scope and characteristics of their pretraining data. For task adaptation, we retain the pretrained encoder parameters and train a sequence-classification head for sentiment prediction.

4.1 Pretrained Encoder Models

Table 1 presents the configurations of Arabic language pretraining. AraBERT provides a general Arabic pretrained encoder and serves as a reference point for evaluating models developed with stronger exposure to dialectal Arabic (Antoun et al., 2020). MARBERTv2 builds on pretraining that incorporates large-scale Arabic social-media content and substantial dialectal variation, making it relevant to sentiment analysis of informal and dialectal text (Abdul-Mageed et al., 2021). In contrast, DarijaBERT is specifically designed for Darija and represents a dialect-specialized pretraining strategy (Gaanoun et al., 2023). This combination enables us to compare general Arabic, broadly dialect-oriented, and dialect-specific representations within the same multi-dialect sentiment classification task.



(a) Train Set

(b) Test Set

Figure 1: Distribution of samples across the five Arabic dialects and three sentiment classes in the dataset. The test set contains an equal number of samples for each dialect, while the training set has no Lebanese samples.

Table 2: Hyperparameter settings

Hyperparameter	PLM / Value
Learning rate	2e−5
Batch size	16
Epochs	10
Optimizer	AdamW
Warmup	10% of steps
Max sequence length	512
Seed	42

4.2 Sequence Classification and Parameter-Efficient Adaptation

For each encoder, we attach a sequence-classification head that maps the sentence representation to the three sentiment labels: positive, negative, and neutral. During adaptation, the pretrained encoder parameters are kept frozen, and only the parameters of the classification head are updated. This head-only adaptation substantially reduces the number of trainable parameters while preserving the representations learned during pretraining.

The same adaptation strategy is applied to MARBERTv2, DarijaBERT, and AraBERT, ensuring a consistent downstream architecture across all systems. Consequently, differences in performance primarily reflect the representations provided by the respective pretrained encoders rather than differences in the classification mechanism.

5 Experimental Setup

We use the official training and test splits provided by the shared-task organizers. Each input sentence is processed with the tokenizer associated with its corresponding pretrained encoder, and no additional preprocessing is applied. The training hyperparameters are kept consistent across all evaluated models and are summarized in Table 2.

The system is implemented with PyTorch Lightning (Falcon and The PyTorch Lightning team, 2019), while Weights & Biases (W&B) is used for experiment monitoring and logging. Following the official evaluation protocol, Macro F1-Score is used as the primary metric, with Accuracy reported as a complementary measure of overall classification performance.

6 Results

Table 3 reports our three systems' performances alongside the leading submissions on the official leaderboard. Among our models, MARBERTv2 achieves the best performance, with a Macro F1 of 80.11% and Accuracy of 81.90%, placing our submission fifth overall in Subtask 1. AraBERT obtains a Macro F1 of 75.52%, while DarijaBERT reaches 72.74%. MARBERTv2 therefore improves over AraBERT and DarijaBERT by 4.59 and 7.37 Macro F1 points, respectively, maintaining this performance ordering across all reported metrics.

The sizable performance differences among the three systems indicate that backbone selection critically impacts the final classification performance. Since the evaluated systems use the same downstream sequence-classification architecture and head-only adaptation strategy, changing the pretrained encoder results in substantial differences in Macro F1 and Accuracy. In particular, MARBERTv2 consistently outperforms the other two backbones, whereas AraBERT occupies an inter-

Table 3: Leaderboard results for DialectSentEval Subtask 1. Macro F1 Score (%) is the primary metric for comparison across all systems. The top four systems are shown in the first group, while our three submissions are presented in the second group. The best-performing system is highlighted in gray.

Team / System	Metric				Rank
	Macro F1	Accuracy	Precision	Recall	
Other Systems					
Momen_Ibrahim	86.67	88.57	86.38	87.10	1
Axiom	82.11	84.38	81.88	82.42	2
Tasneemduridi	81.75	84.19	81.89	81.62	3
Habiba_ayman	81.41	83.43	81.15	81.97	4
Our System					
MARBERTv2	80.11	81.90	79.77	80.93	
DarijaBERT	72.74	75.43	72.51	73.17	5
AraBERT	75.52	77.33	76.12	76.33	

mediate position and DarijaBERT yields the lowest aggregate performance.

Results also show that the broadly dialect-oriented MARBERTv2 performs better in the aggregate evaluation than both the general Arabic AraBERT and the dialect-specialized DarijaBERT. This pattern suggests that exposure to linguistically diverse dialectal Arabic during pretraining provides representations that transfer more effectively to a multi-dialect sentiment classification setting. At the same time, DarijaBERT's specialization in Moroccan Darija does not translate into stronger overall performance across the complete test set. While specialization toward a single dialect may provide less transferable representations when evaluating across several dialectal varieties, we cannot confirm whether DarijaBERT retains an advantage specifically for Moroccan examples without a fine-grained evaluation.

An additional challenge arises from the dataset composition: Lebanese appears in the test set but is absent from the training split, requiring systems to process a dialect unseen during task-specific training. The stronger aggregate performance of MARBERTv2 may therefore associate with greater cross-dialect transfer from its broader pretraining exposure. Nevertheless, aggregate scores alone cannot isolate performance on Lebanese or distinguish unseen-dialect generalization from improvements on dialects observed during training. Because the gold labels for the official test set were not released by the task organizers, our analysis is strictly constrained to these aggregate metrics. Consequently, a fine-grained, per-dialect evaluation to definitively verify these hypotheses regarding cross-dialect transfer and dialect-specific advantages could not be performed on the official test split.

7 Conclusion

This work evaluates three pretrained Arabic encoders—MARBERTv2, DarijaBERT, and AraBERT—on Subtask 1 of the DialectSentEval shared task for multi-dialect sentiment analysis. Formulating the task as three-class sequence classification, we employed a parameter-efficient head-only adaptation strategy, keeping pretrained encoder parameters frozen while optimizing the classification head. Our results demonstrate that MARBERTv2 achieves the highest performance, yielding 80.11% Macro F1 and 81.90% Accuracy, securing fifth place overall. These findings highlight the importance of backbone selection, suggesting that broader, dialect-oriented pretraining provides more transferable representations for multi-dialect sentiment classification.

Despite our system's competitive performance, a primary limitation is the reliance on aggregate metrics. Because organizers did not release gold labels for the official test set, fine-grained per-dialect evaluation could not be conducted. Upon their availability, future research should prioritize per-dialect analysis to better understand each encoder's strengths and weaknesses, particularly regarding zero-shot generalization to unseen dialects like Lebanese. Furthermore, exploring alternative adaptation strategies—such as leveraging decoder-only Large Language Models (LLMs) or incorporating dialect-specific preprocessing—could further enhance performance. Overall, this study contributes to understanding pretrained Arabic models in multi-dialect contexts, establishing a solid foundation for future research.

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